Chapter 18: Smaller Ecdysozoans

Introduction to Ecdysozoa

- **Protostome** animals are divided into two large clades: **Lophotrochozoa** and **Ecdysozoa**.
- **Ecdysozoa** comprises taxa that molt their cuticle as they grow.
- **Cuticle**: A nonliving external layer secreted by the epidermis.
- **Ecdysis**: The process of shedding the outer layer of the cuticle.
- Molting is regulated by the hormone **ecdysone**.
- Ecdysozoan phyla include:
  - **Nematoda** (Roundworms)
  - **Nematomorpha** (Horsehair worms)
  - **Kinorhyncha**
  - **Priapulida**
  - **Loricifera**
  - **Onychophora** (Velvet worms)
  - **Tardigrada** (Water bears)
  - **Arthropoda** (Arthropods)
- Body Plans:
  - **Pseudocoelomate**: Nematoda, Nematomorpha, Kinorhyncha, Priapulida (assumed). Pseudocoelom used as a hydrostatic skeleton.
  - **Acoelomate** or **Pseudocoelomate**: Loricifera (varies).
  - **Coelomate** (reduced coelom): Panarthropoda (Onychophora, Tardigrada, Arthropoda).

Figure 1: Cladogram depicting one hypothesis for relationships among ecdysozoan phyla. Characters shown are subsets of those in Nielsen (1995), Neuhaus and Higgins (2002), and Brusca and Brusca (2003); the nematode character “6 + 6 + 4 sensilla” refers to the anterior rings of sensory papillae.

Phylum Nematoda: Roundworms

- Approximately 25,000 species named; estimates up to 500,000.

- Found in sea, freshwater, soil, and as parasites of plants and animals.
- **Free-living nematodes** feed on bacteria, yeasts, fungal hyphae, algae, or are predatory.
- **Parasitic nematodes** affect virtually every type of animal and many plants.
- **Caenorhabditis elegans**: An important model organism for developmental biology and genetics.

Form and Function

- Cylindrical shape; flexible, nonliving **cuticle**.
- Lack motile cilia or flagella (except in one species).
- **Eutely**: Constant number of cells.
- **Muscles**: Longitudinal only; no circular muscles.
- **Excretory system**: Gland cells or canal system; no protonephridia.
- **Pharynx**: Muscular with a triradiate lumen.
- **Cuticle**: Secreted by the underlying epidermis (**hypodermis**).
  - Composed primarily of **collagen**.
  - Contains high hydrostatic pressure (**turgor**).
  - Shed during juvenile growth stages.
- **Hypodermis**: Syncytial; nuclei located in four **hypodermal cords** (dorsal, ventral, two lateral).
  - Dorsal and ventral cords bear nerves.
  - Lateral cords bear excretory canals.

Figure 2: A, Structure of a nematode as illustrated by **Ascaris** female. **Ascaris** has two ovaries and uteri, which open to the outside by a common genital pore. B, Cross section. C, Single muscle cell; spindle abuts hypodermis, muscle arm extends to dorsal or ventral nerve.

- **Body-wall muscles**:
  - Lie beneath the hypodermis.
  - Contract longitudinally only.
  - Arranged in four bands.
  - Each muscle cell has a contractile **fibrillar portion** (spindle) and a noncontractile **sarcoplasmic portion** (cell body).

    - **Muscle arm**: Process extending from the cell body to the dorsal or ventral nerve (unusual arrangement).
- **Hydrostatic Skeleton**:
    - Fluid-filled **pseudocoel**.
    - Cuticle antagonizes longitudinal muscles (no circular muscles).
    - Contraction compresses the cuticle; relaxation is aided by internal pressure stretching the cuticle.
    - High hydrostatic pressure required for efficiency.
- **Alimentary Canal**:
    - Mouth, muscular pharynx, nonmuscular intestine, short rectum, terminal anus.
    - Food sucked into pharynx; intestine one cell-layer thick.
    - Defecation by muscle pulling anus open; expulsion by pseudocoelomic pressure.
- **Metabolism**:
    - Parasitic adults often anaerobic (glycolysis).
    - Free-living forms often aerobic (Krebs cycle, cytochrome system).
- **Nervous System**:
    - Ring of nerve tissue and ganglia around the pharynx.
    - Dorsal and ventral nerve cords.
    - **Sensory papillae** around head and tail.
    - **Amphids**: Complex sensory organs on the head (chemoreceptive).
    - **Phasmids**: Sensory organs near the posterior end in parasitic nematodes.

Figure 3: Diagram of an amphid in **Caenorhabditis elegans**.

- **Reproduction**:
    - Most are **dioecious**.
    - Males smaller than females; posterior end usually has **copulatory spicules**.
    - Fertilization is internal.
    - Sperm lack flagella and acrosome; amoeboid movement.
    - Development typically direct with four juvenile stages separated by molts.

Figure 4: A, Cross section of a male nematode. B, Posterior end of a male nematode.

Representative Nematode Parasites

Figure 5: Common Parasitic Nematodes of Humans in North America.

Ascaris lumbricoides: The Large Roundworm of Humans

- Common parasite; over 1 billion people infected worldwide.
- **Life Cycle**:
  - Eggs passed in feces; develop in soil (2 weeks).
  - Infection via ingestion of embryonated eggs (contaminated food/soil).
  - Juveniles hatch in intestine, burrow into veins/lymph vessels.
  - Carried to heart and lungs; break into alveoli.
  - Migrate up trachea, swallowed, return to intestine.
  - Mature in intestine (2 months after ingestion).
- Symptoms: Abdominal symptoms, allergic reactions, intestinal blockage, peritonitis.
- **Visceral larva migrans**: Caused by **Toxocara** (dog/cat roundworms) in humans.

Figure 6: A, Intestinal roundworm **Ascaris lumbricoides**, male and female. Male, top, is smaller and has characteristic sharp kink in the end of the tail. Females of this large nematode may be over 30 cm long. B, Intestine of a pig, nearly completely blocked by **Ascaris suum**. Such heavy infections are also fairly common with **A. lumbricoides** in humans.

Hookworms

- **Necator americanus** and **Ancylostoma duodenale**.
- Anterior end curves dorsally (hook-like).
- Large plates in mouth cut intestinal mucosa to suck blood.
- Cause anemia, retarded growth in children.

Figure 7: A, Mouth of hookworm displaying cutting plates. B, Section through anterior end of hookworm attached to dog intestine. Note cutting plates pinching off mucosa from which the thick muscular pharynx sucks blood. Esophageal glands secrete anticoagulant to prevent blood from clotting.

- **Life Cycle**:
  - Eggs passed in feces; juveniles hatch in soil.
  - Infective juveniles burrow through human skin (contact with soil).
  - Migrate to lungs via blood, then to intestine.

Figure 8: The life cycle of hookworms: a shelled embryo develops into a first-stage juvenile which is followed by two molts. The resulting third-stage juvenile enters developmental arrest until it reaches a new host (A to C). Human infection may be via the mouth (D) or skin (E). Juveniles migrate through the circulatory system to lungs (F), enter alveoli (G), and then reach the intestine where they mate (H).

Trichina Worm (Trichinella spiralis)

- Causes **trichinosis**.
- Adult worms in small intestine; females produce live young.
- Juveniles penetrate blood vessels, migrate to skeletal muscle.
- Penetrate muscle cells, redirect gene expression to form **nurse cells**.
- Infection via eating raw/undercooked meat (pork, etc.) containing encysted juveniles.
- Can infect humans, hogs, rats, cats, dogs.

Figure 9: Muscle infected with trichina worm **Trichinella spiralis**. The juveniles lie within muscle cells that the worms have induced to transform into nurse cells (commonly called cysts). An inflammatory reaction occurs around the nurse cells. Juveniles may live 10 to 20 years, and nurse cells eventually may calcify.

Pinworms (Enterobius vermicularis)

- Most common nematode parasite in the U.S.
- Adults in large intestine and cecum.
- Females migrate to anal region at night to lay eggs.
- Causes itching; scratching contaminates hands/bedclothes.
- Infection via ingestion of eggs.
- Diagnosis: “Scotch tape method”.

Figure 10: Pinworms, **Enterobius vermicularis**. A, Female worm from human large intestine (slightly flattened in preparation), magnified about 20 times. B, Group of pinworm eggs, which are usually discharged at night around the anus of the host, who, by scratching during sleep, gets fingernails and clothing contaminated.

Filarial Worms

- **Wuchereria bancrofti** and **Brugia malayi**:
  - Live in lymphatic system.
  - Transmitted by mosquitos.
  - Females release **microfilariae** into blood/lymph.

Figure 11: Life cycle of **Wuchereria bancrofti**: mosquito ingests microfilariae which penetrate mosquito gut wall and develop to infective juveniles. Juveniles escape from mosquito's proboscis when the insect is feeding and then penetrate wound. (A to C, Juveniles migrate to regional lymph nodes and develop to sexual maturity in afferent lymphatic vessels. Adult worms produce microfilariae, which enter blood circulation (D to G).

- Symptoms include inflammation and obstruction of the lymphatic system.
- Long-term obstruction can lead to **elephantiasis** (excessive growth of connective tissue and swelling).

Figure 12: Elephantiasis of leg caused by adult filarial worms of **Wuchereria bancrofti**, which live in lymph passages and block the flow of lymph. Tiny juveniles, called microfilariae, are ingested with blood meal of mosquitos, where they develop to infective stage and are transmitted to a new host.

- **Onchocerca volvulus**: Causes river blindness; transmitted by black flies.
- **Dirofilaria immitis** (Dog heartworm):
  - Infects dogs, cats, etc.
  - Transmitted by mosquitos.
  - Lives in heart and pulmonary arteries.

Figure 13: **Dirofilaria immitis** in right ventricle, extending up into right and left pulmonary arteries of an eight-year-old Irish setter.

Phylum Nematomorpha: Horsehair Worms

- "Horsehair worms"; superficially resemble hairs.
- Sister taxon to Nematoda (Clade **Nematoidea**).
- Adults free-living in moist habitats; juveniles parasitic in arthropods.
- **Form and Function**:
  - Extremely long and slender (up to 1 m).
  - Cuticle, hypodermis, longitudinal muscles only (like nematodes).
  - Vestigial digestive system; absorb nutrients through body wall (adults and larvae).
  - No circulatory, respiratory, or excretory systems.
  - Nerve ring and midventral nerve cord.
- **Life Cycle**:
  - Juveniles encyst on vegetation or in hosts.
  - Parasitize arthropods (e.g., grasshoppers, crabs).
  - Molt once and emerge as adults into water (stimulate terrestrial host to seek water).
  - Dioecious.

Figure 14: Structure of **Paragordius**, a nematomorph. A, Longitudinal section through the anterior end. B, Transverse section. C, Posterior end of male and female worms. Nematomorphs, or "horsehair worms," are very long and very thin. Their pharynx is usually a solid cord of cells and appears nonfunctional. **Paragordius**, whose pharynx opens through to the intestine, is unusual in this respect and also in the possession of a photosensory organ ("eye"). D, **Paragordius tricuspidatus** emerges from the body of a European cricket, **Nemobius sylvestris**.

Phylum Kinorhyncha

- Marine worms, < 1 mm long.
- "Spiny necked".
- **Form and Function**:
  - Body divided into head, neck, and trunk (11 segments).
  - Retractile head (**introvert**) with circlets of spines (**scalids**).
  - Scalids function in locomotion, chemoreception, mechanoreception.
  - Chitinous cuticle, epidermal cords.
  - Circular, longitudinal, and diagonal muscles.
  - Burrow in mud/silt by extending head and anchoring with spines.
  - Complete digestive system; feed on diatoms/organic material.
  - Pseudocoel filled with amebocytes.
  - Protonephridia (solenocytes).
  - Nervous system with brain and ventral nerve cord.
  - Dioecious; 6 juvenile stages.

Figure 15: A, **Echinoderes**, a kinorhynch, is a minute marine worm. Segmentation is superficial. The head, with its circle of spines, is retractile. B, Colored scanning electron micrograph (SEM) of kinorhynch **Antigomonas** sp.

Phylum Priapulida

- Small group of marine worms; cold waters.
- Burrowing predators or detritus feeders.
- **Form and Function**:
  - Cylindrical body: **introvert**, **trunk**, and **caudal appendages**.
  - Retractable introvert with papillae and rows of curved spines (scalids).
  - Trunk superficially segmented (30-100 rings).
  - Caudal appendages: Respiratory/chemoreceptive.
  - Chitinous cuticle (molted).
  - Muscular pharynx, straight intestine.
  - Nerve ring and midventral nerve cord.
  - Hemerythrin respiratory pigment in some.
  - Sexes separate; urogenital organs with solenocytes.

Figure 16: A, Major internal structures of **Priapulus**. B, **Priapulus caudatus** from Lurefjord, Norway.

Phylum Loricifera

- Tiny animals (0.10-0.50 mm) living in marine gravel.
- Protective external case (**lorica**).
- **Form and Function**:
  - Body regions: Mouth cone, head (introvert), neck, thorax, abdomen.
  - Introvert with 9 circlets of scalids.
  - Entire forepart retracts into lorica.
  - Diet unknown (possibly bacteria).
  - Dioecious; distinct larval phase (**Higgins larva**).

Figure 17: A, Dorsal view of adult loriciferan, **Nanaloricus mysticus**, showing internal features. B, Live animal, 0.3 mm.

Clade Panarthropoda

- Contains **Arthropoda**, **Onychophora**, and **Tardigrada**.
- Coelom reduced; **hemocoel** develops (lined by extracellular matrix).
- Open circulatory system; muscular heart.

Phylum Onychophora: Velvet Worms

- “Velvet worms” or “walking worms”.
- Caterpillar-like; tropical/subtropical moist habitats.
- Predaceous (feed on insects, snails, worms).
- Changed little in 500 million years (fossil **Aysheaia**).
- **Form and Function**:
  - Cylindrical, no external segmentation except paired appendages.
  - Soft, velvety skin; thin, flexible cuticle (protein + chitin, molted in patches).
  - Body studded with **tubercles**.
  - Head: Pair of **antennae**, annelid-like eyes, ventral mouth with **mandibles** and **oral papillae**.
  - 14-43 pairs of **unjointed legs** (stubby, clawed).

Figure 18: **Peripatus**, a caterpillar-like onychophoran that has characteristics in common with both annelids and arthropods. A, Ventral view of head. B, In natural habitat.

- **Slime glands**: Expel sticky fluid from oral papillae for defense/prey capture.
- **Tracheal system**: For respiration; spiracles cannot close (restricted to moist habitats).
- **Nephridia**: Pair in each leg-bearing segment.
- Open circulatory system; dorsal tubular heart.
- Nervous system: Ladder-like.
- Reproduction: Dioecious; oviparous, ovoviviparous, or viviparous (some placental).

Figure 19: Internal anatomy of an onychophoran.

Phylum Tardigrada: Water Bears

- “Water bears”; minute (< 1 mm).
- Mostly terrestrial (water film of mosses/lichens), some freshwater/marine.
- **Form and Function**:
  - Elongated, cylindrical, unsegmented body.
  - Four pairs of short, stubby, **unjointed legs** with claws.
  - Nonchitinous cuticle (molted).

Figure 20: Scanning electron micrograph of an aquatic tardigrade, **Pseudobiotus**.

- Mouth with **buccal tube** and **stylets** (piercing/sucking).
- **Malpighian tubules**: Three glands for excretion.
- No circulatory or respiratory systems (diffusion).
- Hemocoel; true coelom restricted to gonadal cavity.
- Brain and double ventral nerve cord.
- **Cryptobiosis**: State of suspended animation to withstand harsh conditions (desiccation, temperature extremes).
  - Water content decreases to 3%; “tun” state.

Figure 21: Internal anatomy of a tardigrade.

- Reproduction: Dioecious (some parthenogenetic). Eggs often ornate.

Figure 22: Scanning electron micrograph of a highly ornate egg of a tardigrade, **Macrobiotus hufelandii**.

- Molting occurs periodically; females of some species deposit eggs in the molted cuticle.

Figure 23: Molted cuticle of a tardigrade, containing a number of fertilized eggs.

Phylogeny

- **Ecdysozoa**: United by molting of cuticle.
- **Nematoidea**: Nematoda + Nematomorpha (collagenous cuticle).
- **Scalidophora** (proposed): Kinorhyncha + Priapulida + Loricifera (scalids, introvert).
- **Panarthropoda**: Onychophora + Tardigrada + Arthropoda.
  - Onychophora sister to Arthropoda + Tardigrada.
  - Onychophora share with Arthropoda: Tubular heart, hemocoel, tracheae, large brain.
  - Onychophora share with Annelida: Metameric nephridia, muscular body wall, pigment cup ocelli.
  - Tardigrada share with Arthropoda: Arthropod-type setae, muscles inserting on cuticle, DNA sequence support.